

CV Equations

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Legend

- State 1:
 - Three inlets/injectors: 2 smaller ones for air and 1 for hydrogen
 - Subscript “A” stands for air
 - Subscript “B” stands for hydrogen
 - ρ (kg/m^3) - density
 - u (m/s) – velocity
 - A_A (m^2) - area of the two air injectors
 - A_B (m^2) - area of the hydrogen injector
 - h (J/kg) – static enthalpy – it is a function of T
 - $\dot{m}_A$ (kg/s) - mass flow rate out of air injectors
 - $\dot{m}_B$ (kg/s) - mass flow rate out of hydrogen injectors
- State 2:
 - One outlet (the inlet of the preburner)
 - Subscript “i” stands for
 - ρ (kg/m^3) - density
 - u (m/s) – velocity
 - A_i (m^2) – area of preburner inlet
 - h (J/kg) – static enthalpy – it is a function of T
 - $\dot{m}_i$ (kg/s) - mass flow rate at preburner inlet

Main equations

- Mass Conservation

- $\dot{m}_A + \dot{m}_B = \dot{m}_i$
- $\dot{m}_A + \dot{m}_B - \dot{m}_i = 0$
- $\rho_A * u_A * A_A + \rho_B * u_B * A_B - \rho_i * u_i * A_i = 0$

- Momentum Conservation

- $\int_{CS} \rho * u * u * dA = \sum F$
- $\dot{m}_i * u_i - \sum_{Inlets} (\dot{m} * u) = \sum F$

- Energy Conservation

- $\int_{CS} \rho * \left(h + \frac{u^2}{2} \right) * u * dA = \dot{Q} - \dot{W}$
- Assumptions: Adiabatic ($\dot{Q} = 0$), no shaft work ($\dot{W} = 0$)
- $\int_{CS} \rho * \left(h + \frac{u^2}{2} \right) * u * dA = 0$
- $\dot{m}_i * \left(h_i + \frac{u_i^2}{2} \right) - \sum_{Inlets} (\dot{m} * \left(h + \frac{u^2}{2} \right)) = 0$

Momentum Conservation Equation

- $\dot{m}_i * u_i - \sum_{Inlets} (\dot{m} * u) = \sum F$
- $\dot{m}_i * u_i - \sum_{Inlets} (\dot{m} * u) = (P_A * A_A + P_B * A_B) - P_i * A_i$
- $\dot{m}_i * u_i - \sum_{Inlets} (\dot{m} * u) - (P_A * A_A + P_B * A_B) + P_i * A_i = 0$
- $\dot{m}_i * u_i - (\dot{m}_A * u_A + \dot{m}_B * u_B) - (P_A * A_A + P_B * A_B) + P_i * A_i = 0$

Energy Conservation Equation

- $\int_{CS} \rho * \left(h_{static} + \frac{u^2}{2} \right) * u * dA = 0$
- $\dot{m}_i * \left(h_i + \frac{u_i^2}{2} \right) - \sum_{Inlets} (\dot{m} * \left(h + \frac{u^2}{2} \right)) = 0$
- $\dot{m}_i * \left(h_i + \frac{u_i^2}{2} \right) - [(\dot{m}_A * \left(h_A + \frac{u_A^2}{2} \right)) + (\dot{m}_B * \left(h_B + \frac{u_B^2}{2} \right))] = 0$

Equation List

- $F_1 = \rho_A * u_A * A_A + \rho_B * u_B * A_B - \rho_i * u_i * A_i = 0$
- $F_2 = \dot{m}_i * u_i - (\dot{m}_A * u_A + \dot{m}_B * u_B) - (P_A * A_A + P_B * A_B) + P_i * A_i = 0$
- $F_3 = \dot{m}_i * (h_i + \frac{u_i^2}{2}) - [(\dot{m}_A * (h_A + \frac{u_A^2}{2})) + (\dot{m}_B * (h_B + \frac{u_B^2}{2}))] = 0$

Simplifying: From three equations three unknowns to two equations two unknowns.

Part 1:

- $F_1 = \rho_A * u_A * A_A + \rho_B * u_B * A_B = \rho_i * u_i * A_i$
- $F_1 = \dot{m}_A + \dot{m}_B = \dot{m}_i$
- $\frac{F_2}{F_1} = \frac{\dot{m}_i * u_i - (\dot{m}_A * u_A + \dot{m}_B * u_B) - (P_A * A_A + P_B * A_B) + P_i * A_i}{\dot{m}_i} = \frac{0}{\dot{m}_A + \dot{m}_B}$
- $\frac{F_2}{F_1} = u_i - \frac{\dot{m}_A}{\dot{m}_i} * u_A - \frac{\dot{m}_B}{\dot{m}_i} * u_B - \frac{P_A * A_A}{\dot{m}_i} - \frac{P_B * A_B}{\dot{m}_i} + \frac{P_i * A_i}{\dot{m}_i} = 0$
- $P_A = \rho_A * R * T_A, P_B = \rho_B * R * T_B, P_i = \rho_i * R * T_i$
- $\frac{F_2}{F_1} = u_i - \frac{\dot{m}_A}{\dot{m}_i} * u_A - \frac{\dot{m}_B}{\dot{m}_i} * u_B - \frac{\rho_A * R * T_A * A_A}{\dot{m}_i} - \frac{\rho_B * R * T_B * A_B}{\dot{m}_i} + \frac{\rho_i * R * T_i * A_i}{\dot{m}_i} = 0$
- $\rho_A = \frac{\dot{m}_A}{u_A * A_A}, \rho_B = \frac{\dot{m}_B}{u_B * A_B}, \rho_i = \frac{\dot{m}_i}{u_i * A_i}$
- $\frac{F_2}{F_1} = u_i - \frac{\dot{m}_A}{\dot{m}_i} * u_A - \frac{\dot{m}_B}{\dot{m}_i} * u_B - \frac{\frac{\dot{m}_A}{u_A * A_A} * R * T_A * A_A}{\dot{m}_i} - \frac{\frac{\dot{m}_B}{u_B * A_B} * R * T_B * A_B}{\dot{m}_i} + \frac{\frac{\dot{m}_i}{u_i * A_i} * R * T_i * A_i}{\dot{m}_i} = 0$
- $\frac{F_2}{F_1} = u_i - \frac{\dot{m}_A}{\dot{m}_i} * u_A - \frac{\dot{m}_B}{\dot{m}_i} * u_B - \frac{\dot{m}_A * R * T_A}{\dot{m}_i * u_A} - \frac{\dot{m}_B * R * T_B}{\dot{m}_i * u_B} + \frac{R * T_i}{u_i} = 0$
- $E_1 = u_i - \frac{\dot{m}_A}{\dot{m}_i} * u_A - \frac{\dot{m}_B}{\dot{m}_i} * u_B - \frac{\dot{m}_A * R * T_A}{\dot{m}_i * u_A} - \frac{\dot{m}_B * R * T_B}{\dot{m}_i * u_B} + \frac{R * T_i}{u_i} = 0$

Simplifying: From three equations three unknowns to two equations two unknowns.

Part 2:

- $F_1 = \rho_A * u_A * A_A + \rho_B * u_B * A_B = \rho_i * u_i * A_i$

- $F_1 = \dot{m}_A + \dot{m}_B = \dot{m}_i$

- $\frac{F_3}{F_1} = \frac{\dot{m}_i * (h_i + \frac{u_i^2}{2}) - [(\dot{m}_A * (h_A + \frac{u_A^2}{2})) + (\dot{m}_B * (h_B + \frac{u_B^2}{2}))]}{\dot{m}_i} = \frac{0}{\dot{m}_A + \dot{m}_B}$

- $\frac{F_3}{F_1} = (h_i + \frac{u_i^2}{2}) - \frac{\dot{m}_A}{\dot{m}_i} (h_A + \frac{u_A^2}{2}) - \frac{\dot{m}_B}{\dot{m}_i} (h_B + \frac{u_B^2}{2}) = 0$

- $E_2 = (h_i + \frac{u_i^2}{2}) - \frac{\dot{m}_A}{\dot{m}_i} (h_A + \frac{u_A^2}{2}) - \frac{\dot{m}_B}{\dot{m}_i} (h_B + \frac{u_B^2}{2}) = 0$

Final Equations:

- $E_1 = u_i - \frac{\dot{m}_A}{\dot{m}_i} * u_A - \frac{\dot{m}_B}{\dot{m}_i} * u_B - \frac{\dot{m}_A * R * T_A}{\dot{m}_i * u_A} - \frac{\dot{m}_B * R * T_B}{\dot{m}_i * u_B} + \frac{R * T_i}{u_i} = 0$
- $E_2 = (h_i + \frac{u_i^2}{2}) - \frac{\dot{m}_A}{\dot{m}_i} (h_A + \frac{u_A^2}{2}) - \frac{\dot{m}_B}{\dot{m}_i} (h_B + \frac{u_B^2}{2}) = 0$

Equations for adjusting state 1 then resolving mixing. Injector A and B

- $\dot{S}_{gen} = \dot{m}_A * (s_{A1_2} - s_{A1})$
- $P_{oA1} = P_A \left(1 + \frac{\gamma(T_A)-1}{2} * M_A^2 \right)^{\frac{\gamma(T_A)}{\gamma(T_A)-1}}$ **state 1 pre mixing**
- $P_{oA1_2} = e^{-\frac{\dot{S}_{gen}}{R}} * P_{oA1}$ **oA1_2 means that a Sgen has been introduced**
- $P_{oA1_2} = e^{-\frac{\dot{S}_{gen}}{R}} * P_{oA1}$
- $P_{oA1_2} = P_{A1_2} \left(1 + \frac{\gamma(T_{A1_2})-1}{2} * M_{A1_2}^2 \right)^{\frac{\gamma(T_{A1_2})}{\gamma(T_{A1_2})-1}}$
- $P_{A1_2} = \frac{P_{oA1_2}}{\left(1 + \frac{\gamma(T_{A1_2})-1}{2} * M_{A1_2}^2 \right)^{\frac{\gamma(T_{A1_2})}{\gamma(T_{A1_2})-1}}}$
- $T_{oA1} = T_{A1} \left(1 + \frac{\gamma(T_{A1})-1}{2} * M_{A1}^2 \right) = T_{oA1_2} = T_{A1_2} \left(1 + \frac{\gamma(T_{A1_2})-1}{2} * M_{A1_2}^2 \right)$
- $T_{A1_2} = \frac{T_{oA1}}{\left(1 + \frac{\gamma(T_{A1})-1}{2} * M_{A1}^2 \right)}$

Equations for adjusting state 1 then resolving mixing. Injector A and B cont.

- $\rho_{A1_2} = \frac{P_{A1_2}}{R * T_{A1_2}}$
- $\rho_{A1_2} = \frac{P_{oA1_2} * \left(1 + \frac{\gamma(T_{A1_2}) - 1}{2} * M_{A1_2}^2\right)}{\left(1 + \frac{\gamma(T_{A1_2}) - 1}{2} * M_{A1_2}^2\right) \frac{\gamma(T_{A1_2})}{\gamma(T_{A1_2}) - 1} * R * T_{oA1}}$
- $\rho_{A1_2} = \frac{P_{oA1_2}}{R * T_{oA1}} * \frac{\left(1 + \frac{\gamma(T_{A1_2}) - 1}{2} * M_{A1_2}^2\right)}{\left(1 + \frac{\gamma(T_{A1_2}) - 1}{2} * M_{A1_2}^2\right) \frac{\gamma(T_{A1_2})}{\gamma(T_{A1_2}) - 1}}$
- $\rho_{A1_2} = \frac{P_{oA1_2}}{R * T_{oA1}} * \left(1 + \frac{\gamma(T_{A1_2}) - 1}{2} * M_{A1_2}^2\right) * \left(1 + \frac{\gamma(T_{A1_2}) - 1}{2} * M_{A1_2}^2\right)^{-1 * \left(\frac{\gamma(T_{A1_2})}{\gamma(T_{A1_2}) - 1}\right)}$
- $\rho_{A1_2} = \frac{P_{oA1_2}}{R * T_{oA1}} * \left(1 + \frac{\gamma(T_{A1_2}) - 1}{2} * M_{A1_2}^2\right)^{1 - 1 * \left(\frac{\gamma(T_{A1_2})}{\gamma(T_{A1_2}) - 1}\right)}$
- $\rho_{A1_2} = \frac{P_{oA1_2}}{R * T_{oA1}} * \left(1 + \frac{\gamma(T_{A1_2}) - 1}{2} * \frac{u_{A1_2}^2}{\gamma(T_{A1_2}) * R * T_{A1_2}}\right)^{1 - 1 * \left(\frac{\gamma(T_{A1_2})}{\gamma(T_{A1_2}) - 1}\right)}$

Equations for adjusting state 1 then resolving mixing. Injector A and B cont.

- $\dot{m}_A = \rho_{A1_2} * u_{A1_2} * A_A$
- $\dot{m}_A = \left(\frac{P_{oA1_2}}{R * T_{oA1}} * \left(1 + \frac{\gamma(T_{A12}) - 1}{2} * \frac{u_{A12}^2}{\gamma(T_{A12}) * R * T_{A12}} \right)^{1 - 1 * \left(\frac{\gamma(T_{A12})}{\gamma(T_{A12}) - 1} \right)} \right) * u_{A1_2} * A_A$
- $E_3 = \left(\frac{P_{oA1_2}}{R * T_{oA1}} * \left(1 + \frac{\gamma(T_{A12}) - 1}{2} * \frac{u_{A12}^2}{\gamma(T_{A12}) * R * T_{A12}} \right)^{1 - 1 * \left(\frac{\gamma(T_{A12})}{\gamma(T_{A12}) - 1} \right)} \right) * u_{A12} * A_A - \dot{m}_A$
- $E_4 = \left(\frac{P_{oB1_2}}{R * T_{oB1}} * \left(1 + \frac{\gamma(T_{B12}) - 1}{2} * \frac{u_{B12}^2}{\gamma(T_{B12}) * R * T_{B12}} \right)^{1 - 1 * \left(\frac{\gamma(T_{B12})}{\gamma(T_{B12}) - 1} \right)} \right) * u_{B12} * A_B - \dot{m}_B$

Equations to for the injectors A and B

- $T_{oA1} = T_{A1_2} \left(1 + \frac{\gamma(T_{A1_2})-1}{2} * \frac{u_{A1_2}^2}{\gamma(T_{A1_2}) * R * T_{A1_2}} \right)$
- $E_5 = T_{A1_2} \left(1 + \frac{\gamma(T_{A1_2})-1}{2} * \frac{u_{A1_2}^2}{\gamma(T_{A1_2}) * R * T_{A1_2}} \right) - T_{oA1}$
- $E_6 = T_{B1_2} \left(1 + \frac{\gamma(T_{B1_2})-1}{2} * \frac{u_{B1_2}^2}{\gamma(T_{B1_2}) * R * T_{B1_2}} \right) - T_{oB1}$